SE444: Introduction to Artificial Intelligence
Makeup Mid-Term II Fall 2025 – Solutions

Prof. Anis Koubaa
akoubaa@alfaisal.edu
Alfaisal University

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Part I. Critical-Thinking Multiple Choice Questions

Total Score: 4 points **Total Time:** 20 minutes

Instructions: Each question may have **one or more correct answers**. Select **all that apply**. Partial credit will be awarded for partially correct answers.

Answer Key Summary

Option	Q1 (5m)	Q2 (5m)	Q3 (5m)	Q4 (5m)
A	**X**		**X**	
B			**X**	**X**
C	**X**	**X**	**X**	
D				**X**
E			**X**	

Correct Answers:

- **Q1 (FOL):** A, C
- **Q2 (Propositional Logic):** C
- **Q3 (CSP):** A, B, C, E
- **Q4 (Agents):** B, D

Question I.1: First-Order Logic (FOL) — Chained Implications

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

You are given the following FOL sentences (assume a non-empty domain):

1. $\forall x\,(R(x) \to S(x))$
2. $\forall x\,(S(x) \to T(x))$
3. $\exists x\,R(x)$

Which of the following sentences are logically entailed by (1), (2), and (3)?

A) $\exists x\,T(x)$

B) $\forall x\,T(x)$

C) $\exists x\,(R(x) \wedge T(x))$

D) $\forall x\,(T(x) \to R(x))$

E) $\exists x\,(S(x) \wedge \neg R(x))$

Answer: A and C

Explanation: From (3), there exists some a such that $R(a)$. By (1), $R(a) \to S(a)$, so $S(a)$ holds. By (2), $S(a) \to T(a)$, so $T(a)$ holds. This gives us the chain $R(a) \to S(a) \to T(a)$.

Details:

A) **Correct.** Since $R(a)$ implies $S(a)$ implies $T(a)$, we have $\exists x\,T(x)$.

B) **Incorrect.** The premises only guarantee that objects satisfying R also satisfy T. They don't say anything about objects that don't satisfy R.

C) **Correct.** The same a that satisfies $R(a)$ also satisfies $T(a)$ through the chain. Thus, $\exists x\,(R(x) \wedge T(x))$ is true.

D) **Incorrect.** The chain goes $R \to S \to T$, not the reverse. There could be objects satisfying T without satisfying R.

E) **Incorrect.** There is no information ensuring the existence of an element that satisfies $S(x)$ but not $R(x)$. Such elements might exist, but are not entailed.

Key Concept: Chained universal implications with an existential premise allow inference of existence claims that follow the chain, but not universal claims or reverse implications.

Question I.2: Propositional Logic — Satisfiability Analysis

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

Consider a knowledge base (KB) with propositional atoms a, b, c, d, e:

1. $(a \wedge b) \rightarrow c$
2. $c \rightarrow d$
3. $d \rightarrow e$
4. $\neg e$
5. b

Which of the following statements is true?

A) KB is unsatisfiable.

B) KB $\models a$.

C) KB $\models \neg a$.

D) KB $\models c$.

E) KB can be satisfied with $a =$ true and $c =$ true.

Answer: C

Explanation: The KB is satisfiable but forces a to be false through backward reasoning.

Step-by-step analysis:

Step 1: From (4), we know $\neg e$, so e is false.

Step 2: From (3), $d \rightarrow e$. Since e is false, d must be false (otherwise $d \rightarrow e$ would be false).

Step 3: From (2), $c \rightarrow d$. Since d is false, c must be false.

Step 4: From (5), we know b is true.

Step 5: From (1), $(a \wedge b) \rightarrow c$. We know b is true and c is false. For this implication to hold, $(a \wedge b)$ must be false. Since b is true, we need a to be false.

Verification: With $a = F, b = T, c = F, d = F, e = F$:

- (1): $(F \wedge T) \rightarrow F = F \rightarrow F = T$
- (2): $F \rightarrow F = T$
- (3): $F \rightarrow F = T$
- (4): $\neg F = T$
- (5): T

Evaluation of options:

A) **Incorrect.** KB is satisfiable as shown above.

B) **Incorrect.** KB forces a to be false, not true.

C) **Correct.** KB $\models \neg a$ because a must be false in all models.

D) **Incorrect.** KB forces c to be false.

E) **Incorrect.** If $c = T$, then from (2) $d = T$, and from (3) $e = T$. But (4) requires $\neg e$. Contradiction.

Key Concept: Backward chaining from a negated conclusion propagates constraints through implications, forcing specific truth values.

Question I.3: Constraint Satisfaction Problems — Arc Consistency

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 2

Consider a CSP with variables X, Y, Z.
Initial Domains: $D(X) = \{1, 2, 3\}$, $D(Y) = \{2\}$, $D(Z) = \{1, 2, 3\}$
Constraints:

- $X \neq Y$
- $Y < Z$
- $X < Z$

Which are true?

A) After AC, $D(Z)$ becomes $\{3\}$.

B) After AC, $D(X)$ becomes $\{1\}$.

C) After AC, all domains become singletons and the CSP is solved without search.

D) After AC, $D(X)$ still contains the value 2.

E) If we change the constraint "$Y < Z$" to "$Y \leq Z$", then $D(Z)$ would include $\{2, 3\}$ after AC.

Answer: A, B, C, E
Explanation: Arc consistency propagates the singleton domain $D(Y) = \{2\}$ through all constraints.
Details:

A) **Correct.** From $Y < Z$ with $Y = 2$, we need $Z > 2$. From $D(Z) = \{1, 2, 3\}$, only $Z = 3$ satisfies this. AC prunes to $D(Z) = \{3\}$.

B) **Correct.** From $X \neq Y$ with $Y = 2$, we get $X \neq 2$. From $X < Z$ with $Z = 3$, we need $X < 3$, so $X \in \{1, 2\}$. Combining: $X \neq 2$ and $X < 3$ means $X = 1$. So $D(X) = \{1\}$.

C) **Correct.** After AC:

- $D(X) = \{1\}$

- $D(Y) = \{2\}$
- $D(Z) = \{3\}$

All domains are singletons. The assignment $X = 1, Y = 2, Z = 3$ satisfies all constraints. CSP solved without search.

D) **Incorrect.** As shown above, $D(X) = \{1\}$ after AC. Value 2 is removed because $X \neq Y$ requires $X \neq 2$.

E) **Correct.** If "$Y < Z$" becomes "$Y \leq Z$", then with $Y = 2$, we need $Z \geq 2$. From $D(Z) = \{1, 2, 3\}$, values 2 and 3 satisfy this. So $D(Z) = \{2, 3\}$ after this constraint.

Key Concept: When one domain is a singleton, arc consistency can propagate strongly, potentially solving the CSP without backtracking.

Question I.4: Logical (Knowledge-Based) Agents — Environment Analysis

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 3

A hospital deploys a **medical diagnosis robot** operating in a complex healthcare environment that is **partially observable** (limited access to complete patient history), **stochastic** (uncertain test results and treatment responses), **sequential** (each diagnosis affects treatment decisions), and **dynamic** (patient conditions change over time).

A) A simple reflex agent mapping current symptoms directly to diagnoses would be sufficient...

B) A model-based reflex agent could track a patient's medical history...

C) A goal-based agent cannot operate rationally in this environment because treatment outcomes are uncertain...

D) A utility-based agent could weigh the expected benefits and risks of different diagnostic tests...

E) A learning agent would not provide any advantage over a pre-programmed agent in this domain...

Answer: B, D

Explanation: Model-based agents handle partial observability, and utility-based agents can reason about uncertain outcomes and value of information.

Details:

A) **Incorrect.** A simple reflex agent only responds to current percepts without memory. In a partially observable, dynamic environment, the same symptoms could indicate different conditions depending on patient history. A memoryless agent cannot handle this complexity.

B) **Correct.** A model-based reflex agent maintains internal state representing beliefs about

unobserved aspects (prior diagnoses, previous test results, medication history). This enables more accurate diagnosis than a simple reflex agent that ignores history.

C) **Incorrect.** Goal-based agents can operate rationally in stochastic environments by reasoning about expected outcomes, using probabilistic models, and planning under uncertainty. Uncertainty doesn't preclude rational goal pursuit.

D) **Correct.** A utility-based agent maximizes expected utility, which can include the value of diagnostic information. It can rationally order additional tests when the expected information gain outweighs the cost (time, money, patient discomfort).

E) **Incorrect.** Medical knowledge evolves (new diseases, treatments, drug interactions). A learning agent can adapt to new patterns, personalize diagnoses based on outcomes, and improve over time. This provides significant advantage over static agents.

Key Concept: Agent architecture must match environment characteristics. Partial observability requires state maintenance, stochasticity requires probabilistic reasoning, and dynamic domains benefit from learning.

Part II. Logical Reasoning in Healthcare Context

Total Score: 10 points **Total Time:** 40 minutes

Given Predicates:

Predicate	Meaning
Doctor(x)	x is a doctor
Patient(x)	x is a patient
Hospital(x)	x is a hospital
WorksAt(d, h)	Doctor d works at hospital h
TreatedBy(p, d)	Patient p is treated by doctor d
HasDisease(p, dis)	Patient p has disease dis
Disease(dis)	dis is a disease
Prescribes(d, p, m)	Doctor d prescribes medication m to patient p
Medication(m)	m is a medication
Treats(m, dis)	Medication m treats disease dis
Examined(d, p)	Doctor d has examined patient p

Question II.1: Translation to FOL

Time: 10 minutes **Score:** 3 points **Mapped CLOs:** CLO 1

Translate the following statement into first-order logic (FOL):
"Every patient who is treated by a doctor at a hospital is prescribed at least one medication that treats one of their diseases."

Answer:
$\forall p, d, h\,(\text{Patient}(p) \wedge \text{Doctor}(d) \wedge \text{Hospital}(h) \wedge \text{TreatedBy}(p, d) \wedge \text{WorksAt}(d, h)) \rightarrow$
$(\exists m, dis\,(\text{Medication}(m) \wedge \text{Disease}(dis) \wedge \text{Prescribes}(d, p, m) \wedge \text{HasDisease}(p, dis) \wedge \text{Treats}(m, dis)))$
Explanation:

- The universal quantifiers $\forall p, d, h$ range over all possible patients, doctors, and hospitals.
- The antecedent ensures we're considering a patient being treated by a doctor who works at a hospital.
- The consequent states there exists at least one medication m and disease dis such that the medication is prescribed to the patient, the patient has the disease, and the medication treats the disease.

Grading Rubric (3 points):

- **1.0 pt:** Correct universal quantifiers and antecedent structure

- **1.0 pt:** Correct existential quantification in consequent
- **1.0 pt:** All predicates used correctly with proper arguments

Question II.2: Translation with Negation

Time: 10 minutes **Score:** 3 points **Mapped CLOs:** CLO 1

Translate the following statement into first-order logic (FOL):
"No doctor prescribes medication to a patient they have not examined."

Answer (any one of the following is correct):

(A) Implication form (prescription requires examination):

$$\forall d\,\forall p\,\forall m\,\big((\text{Doctor}(d) \land \text{Patient}(p) \land \text{Medication}(m) \land \text{Prescribes}(d,p,m)) \rightarrow \text{Examined}(d,p)\big).$$

(B) Contrapositive style (closest to "No doctor prescribes ... not examined"):

$$\forall d\,\forall p\,\forall m\,\big(\text{Doctor}(d) \land \text{Patient}(p) \land \text{Medication}(m) \land \neg\text{Examined}(d,p) \rightarrow \neg\text{Prescribes}(d,p,m)\big).$$

(C) Negated-existence of a violation:

$$\neg\exists d\,\exists p\,\exists m\,\big(\text{Doctor}(d) \land \text{Patient}(p) \land \text{Medication}(m) \land \text{Prescribes}(d,p,m) \land \neg\text{Examined}(d,p)\big).$$

Explanation (brief):

- (A) expresses that prescription implies prior examination.
- (B) says if the patient is not examined, no prescription can occur.
- (C) forbids the existence of any prescription violating the examination rule.

Grading Rubric (3 points):

- **3 pts:** Any logically equivalent correct form
- **2 pts:** Minor errors in quantification or predicate usage
- **1 pt:** Correct general idea but significant errors

Question II.3: Negation and Conjunctive Normal Form (CNF)

Time: 20 minutes **Score:** 4 points **Mapped CLOs:** CLO 1

Negate the following statement and convert it step-by-step into Conjunctive Normal Form (CNF). Show all transformations clearly and justify each step using the appropriate logical rule.

$$\forall x\,(\text{Doctor}(x) \rightarrow \exists y\,(\text{Patient}(y) \land \text{TreatedBy}(y,x)))$$

Solution Steps:

Step 1: Negate the statement
$\neg\forall x\,(\text{Doctor}(x) \rightarrow \exists y\,(\text{Patient}(y) \wedge \text{TreatedBy}(y, x)))$
Rule: We want to prove the negation

Step 2: Move negation inward ($\neg\forall \equiv \exists\neg$)
$\exists x\,\neg(\text{Doctor}(x) \rightarrow \exists y\,(\text{Patient}(y) \wedge \text{TreatedBy}(y, x)))$

Step 3: Remove implication ($A \rightarrow B \equiv \neg A \vee B$)
$\exists x\,\neg(\neg\text{Doctor}(x) \vee \exists y\,(\text{Patient}(y) \wedge \text{TreatedBy}(y, x)))$

Step 4: Apply De Morgan's law
$\exists x\,(\text{Doctor}(x) \wedge \neg\exists y\,(\text{Patient}(y) \wedge \text{TreatedBy}(y, x)))$

Step 5: Move negation inward ($\neg\exists \equiv \forall\neg$)
$\exists x\,(\text{Doctor}(x) \wedge \forall y\,\neg(\text{Patient}(y) \wedge \text{TreatedBy}(y, x)))$

Step 6: Apply De Morgan's law to inner conjunction
$\exists x\,(\text{Doctor}(x) \wedge \forall y\,(\neg\text{Patient}(y) \vee \neg\text{TreatedBy}(y, x)))$

Step 7: Skolemization (replace $\exists x$ with constant a)
$\text{Doctor}(a) \wedge \forall y\,(\neg\text{Patient}(y) \vee \neg\text{TreatedBy}(y, a))$

Step 8: Drop universal quantifier (implicit for CNF)
$\text{Doctor}(a) \wedge (\neg\text{Patient}(y) \vee \neg\text{TreatedBy}(y, a))$

Final CNF (clauses): $\{\text{Doctor}(a)\}$, $\{\neg\text{Patient}(y), \neg\text{TreatedBy}(y, a)\}$

Interpretation: There is a doctor a who treats no patients.

Grading Rubric (4 pts): 1pt negation+quantifier; 1pt implication+DeMorgan; 1pt $\neg\exists$+inner DeMorgan; 1pt Skolem+CNF.

Part III. Constraint Satisfaction Problem (CSP)

Total Score: 6 points **Total Time:** 30 minutes

Part III: Constraint Satisfaction Problems (CSP)

Problem Specification

Context: You are organizing the schedule for **AI Conference presentation sessions** on Wednesday. There are three presentation rooms and three keynote speakers available. Each speaker can present in at most one room at a time, and some sessions overlap in time. Your goal is to assign each room to a speaker in a way that satisfies all constraints.

Rooms and Session Times

Room	Topic	Time
Room 1	AI Ethics	10:00–11:00 am
Room 2	Machine Learning	10:30–11:30 am
Room 3	Robotics	11:00–12:00 pm

Speaker Availability

Speaker	Available for Rooms
Dr. Smith	Rooms 1 and 2
Dr. Jones	Rooms 2 and 3
Dr. Brown	Rooms 1, 2, and 3

Question III.1: Formulate as a CSP

Time: 10 minutes **Score:** 2 points **Mapped CLOs:** CLO 2

Answer:

Variables (one per room):

- R_1, R_2, R_3

Domains (based on speaker availability):

- $D(R_1) = \{\text{Smith}, \text{Brown}\}$
- $D(R_2) = \{\text{Smith}, \text{Jones}, \text{Brown}\}$
- $D(R_3) = \{\text{Jones}, \text{Brown}\}$

Constraints (overlapping sessions cannot share the same speaker):

- $R_1 \neq R_2$ (Rooms 1 and 2 overlap during 10:30–11:00am)
- $R_2 \neq R_3$ (Rooms 2 and 3 overlap during 11:00–11:30am)

Note: Room 1 and Room 3 do not overlap (Room 1 ends at 11:00, Room 3 starts at 11:00), so no constraint is needed between them.

Grading Rubric (2 points):

- **0.5 pt:** Correct variables definition
- **1.0 pt:** Correct domains based on speaker availability
- **0.5 pt:** Correct constraints identifying overlapping sessions

Question III.2: Arc Consistency (AC-3)

Time: 15 minutes **Score:** 3 points **Mapped CLOs:** CLO 2

Answer:

Initial domains:

- $D(R_1) = \{\text{Smith}, \text{Brown}\}$
- $D(R_2) = \{\text{Smith}, \text{Jones}, \text{Brown}\}$
- $D(R_3) = \{\text{Jones}, \text{Brown}\}$

Arc Consistency Process:
Step 1: Check arc (R_2, R_1) for constraint $R_1 \neq R_2$

- For each value in $D(R_2)$, check if there exists a supporting value in $D(R_1)$:
  - $R_2 = \text{Smith}$: R_1 can be Brown (since Smith $\neq$ Brown). **Supported.**
  - $R_2 = \text{Jones}$: R_1 can be Smith or Brown. **Supported.**
  - $R_2 = \text{Brown}$: R_1 can be Smith (since Brown $\neq$ Smith). **Supported.**
- **No pruning** from $D(R_2)$.

Step 2: Check arc (R_1, R_2) for constraint $R_1 \neq R_2$

- For each value in $D(R_1)$, check if there exists a supporting value in $D(R_2)$:
  - $R_1 = \text{Smith}$: R_2 can be Jones or Brown. **Supported.**
  - $R_1 = \text{Brown}$: R_2 can be Smith or Jones. **Supported.**
- **No pruning** from $D(R_1)$.

Step 3: Check arc (R_2, R_3) for constraint $R_2 \neq R_3$

- For each value in $D(R_2) = \{\text{Smith}, \text{Jones}, \text{Brown}\}$, check support in $D(R_3) = \{\text{Jones}, \text{Brown}\}$:
  - $R_2 = \text{Smith}$: R_3 can be Jones or Brown. **Supported.**

- R_2 = Jones: R_3 can be Brown (since Jones $\neq$ Brown). **Supported.**
- R_2 = Brown: R_3 can be Jones (since Brown $\neq$ Jones). **Supported.**

- **No pruning** from $D(R_2)$.

Step 4: Check arc (R_3, R_2) for constraint $R_2 \neq R_3$

- For each value in $D(R_3) = \{\text{Jones}, \text{Brown}\}$, check support in $D(R_2) = \{\text{Smith}, \text{Jones}, \text{Brown}\}$:
  - R_3 = Jones: R_2 can be Smith or Brown. **Supported.**
  - R_3 = Brown: R_2 can be Smith or Jones. **Supported.**
- **No pruning** from $D(R_3)$.

Final Arc-Consistent Domains:

- $D(R_1) = \{\text{Smith}, \text{Brown}\}$ *(unchanged)*
- $D(R_2) = \{\text{Smith}, \text{Jones}, \text{Brown}\}$ *(unchanged)*
- $D(R_3) = \{\text{Jones}, \text{Brown}\}$ *(unchanged)*

Conclusion: Arc consistency does not prune any values in this CSP. All initial domain values have support. The CSP is arc-consistent but not solved—backtracking search is still required.

Grading Rubric (3 points):

- **1.5 pts:** Correct identification that no pruning occurs
- **1.0 pt:** Showing work (checking at least 2 arcs systematically)
- **0.5 pt:** Correct final domains stated clearly

Partial Credit:

- If student incorrectly prunes values but shows logical reasoning: 0.5–1.5 pts
- If student only states final domains without work: max 0.5 pts

Question III.3: Find a Consistent Solution

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 2

Answer:
Multiple valid solutions exist. Any of the following is acceptable:
Solution 1:

- R_1 = Smith

- R_2 = Jones
- R_3 = Brown

Solution 2:

- R_1 = Smith
- R_2 = Brown
- R_3 = Jones

Solution 3:

- R_1 = Brown
- R_2 = Smith
- R_3 = Jones

Solution 4:

- R_1 = Brown
- R_2 = Jones
- R_3 = Brown

Verification (for Solution 1):

- Domain check: Smith $\in D(R_1)$ ✓, Jones $\in D(R_2)$ ✓, Brown $\in D(R_3)$ ✓
- Constraint $R_1 \neq R_2$: Smith $\neq$ Jones ✓
- Constraint $R_2 \neq R_3$: Jones $\neq$ Brown ✓

All constraints satisfied. ✓

Grading Rubric (1 point):

- **1.0 pt:** Any valid assignment that satisfies all constraints
- **0.0 pt:** Invalid assignment (violates constraints or domains)

Note: Students do not need to verify the solution explicitly—a correct assignment is sufficient.

Answer Key Summary:

- Question III.1: Variables: R_1, R_2, R_3; Domains: $\{\text{Smith}, \text{Brown}\}, \{\text{Smith}, \text{Jones}, \text{Brown}\}, \{\text{Jones}, \text{Brown}\}$ Constraints: $R_1 \neq R_2$, $R_2 \neq R_3$
- Question III.2: All domains remain unchanged (no pruning)

- Question III.3: Any valid assignment (e.g., $R_1 = \text{Smith}, R_2 = \text{Jones}, R_3 = \text{Brown}$)